

Worksheet 7: High Order Thinking and Challenge Patterns (HOTS)

Topic Focus: Complex combinations, deep logic, and multi-step pattern recognition.

1. Find the missing number: 2, 3, 5, 8, 12, __, 23.

- A) 16 B) 17 C) 18 D) 19

Hint: Look at the differences: +1, +2, +3, +4. The next addition should be +5. $12+5 = 17$.

2. Look at this letter sequence: Z, W, T, Q, __. What comes next?

- A) N B) O C) M D) P

Hint: Think of the alphabet backwards and skip 2 letters each time.

3. Palindromic numbers (read same forwards/backwards): 11, 22, 33... 99, 101, 111... Next after 121?

- A) 122 B) 131 C) 141 D) 202

Hint: The first and last digit must be the same. After the 120s, the next valid set is in the 130s.

4. Find the WRONG number in this series: 1, 4, 9, 16, 24, 36, 49.

- A) 16 B) 24 C) 36 D) 49

Hint: The series consists of perfect squares. 24 is not a perfect square (should be 25).

5. Pattern: $1 \times 8 + 1 = 9$; $12 \times 8 + 2 = 98$; $123 \times 8 + 3 = 987$. Result of $1234 \times 8 + 4$?

- A) 9876 B) 9875 C) 9864 D) 9874

Hint: The answer builds a decreasing sequence of digits starting from 9.

6. A plant grows 2 cm day one, 4 cm day two, 6 cm day three... Total height gained after 4 days?

- A) 8 cm B) 12 cm C) 20 cm D) 24 cm

Hint: Add them all up: $2 + 4 + 6 + 8$. Do not just take the 4th day's growth.

7. Rule for this sequence? 2, 5, 11, 23, 47, ...

- A) Multiply 3, subtract 1 B) Multiply 2, add 1 C) Add 3, 6, 12... D) Both B and C are correct

Hint: Try both logic tests. $(2 \times 2) + 1 = 5$. AND gap of 3, gap of 6, gap of 12. Both patterns mathematically describe it perfectly

8. There are 4 prime numbers less than 10 (2, 3, 5, 7). How many prime numbers between 10 and 20?

- A) 3 B) 4 C) 5 D) 6

Hint: List them out: 11, 13, 17, 19.

9. Write out all whole numbers from 1 to 50. How many times do you write the digit '4'?

- A) 5 B) 10 C) 14 D) 15

Hint: Ones place: 4, 14, 24, 34, 44 (five times). Tens place: 40, 41, 42, 43, 44, 45, 46, 47, 48, 49 (ten times). Total = 15.

10. Sequence formula: (Position Number $\times$ 3) - 2. (e.g., Pos 1 = $1 \times 3 - 2 = 1$). What is the 10th number?

- A) 28 B) 30 C) 32 D) 25

Hint: Substitute 10 into the formula: $(10 \times 3) - 2$.

11. Collatz Pattern rule: If even, halve it. If odd, multiply by 3 and add 1. Sequence starts with 6. What are the next 3 numbers?

- A) 3, 10, 5 B) 3, 9, 28 C) 12, 6, 3 D) 3, 4, 2

Hint/solu: 6 is even $\rightarrow 3$. 3 is odd $\rightarrow 3 \times 3 + 1 = 10$. 10 is even $\rightarrow 5$.

12. Find the missing term in the alphanumeric pattern: A1, C4, E9, G16, __

- A) H25 B) I20 C) I25 D) J36

Hint: Letters skip one (A, C, E, G, I). Numbers are perfect squares (1, 4, 9, 16, 25).

13. A frog jumps on lily pads. It can jump 1 pad forward, or leap 2 pads forward. To cross 4 pads, how many different combinations of jumps can it take?

- A) 3 B) 4 C) 5 D) 6

Hint: Jumps can be 1-1-1-1, 1-1-2, 1-2-1, 2-1-1, or 2-2. Notice the total is 5, a Virahanka number (its application)

14. If in a pattern $A+A = B$, and $B+B = C$, and $A=3$, what is C ?

- A) 6 B) 9 C) 12 D) 15

Hint: Substitution. $3+3 = 6$ (B). $6+6 = 12$ (C).

15. Compound Rule: 1, 3, 9, 31... (Multiply by previous prime, then add square of position). (e.g. Pos 2 = $1 \times 2 + 1^2 = 3$). What is the rule simpler form?

- A) It's too complex B) $N \times N$ C) (Previous $\times$ Prime) D) Not meant for a simple formula

Concept: While some patterns look chaotic (like multiplying by primes), mathematics teaches us that breaking them into two separate rules (operations and positions) helps solve them. Answer D highlights that not all patterns form a single simple line.